Pre-Analysis Plan

Hypothesis 1. Having daughters leads to stronger support for gender equality (as measured by attitudes)

That is, denoting X as a measure of support for gender equality,
(X for respondents with daughters) > (X for respondents without daughters)

Dependent variables:

As an outcome variable, we propose to measure effects on two indices of gender equality attitudes (one for 2012 and one for 2018). The index should be composed of variables plh0298_v1, plh0302_v1, plh0308_v1 for 2012 and plh0298_v2, plh0302_v2, plh0308_v2 for 2018. The variables that end in _v1 are on a scale of 1 (fully agree) to 4 (do not agree) with a maximum of gender equal attitudes at 4 for the variables plh298_v1 and plh0302_v1. The variable plh0308_v1 has the most gender equal attitude at 1. So, in the first step, we recode this variable in a way that 4 also marks the most gender equal attitude:

recode plh0308_v1 (1 = 4) (2 = 3) (3 = 2) (4 = 1)

In the next step, we add the values from all three variables and divide them by 12 (the maximum sum) to create a score for gender equality attitudes that ranges from 0 to 1 with 1 being the maximum of gender equality attitudes.

Additionally, an index for gender equality in 2018 is our second outcome variable. For 2018, the variables that end on _v2 are on a scale of 1 (do not agree at all) to 7 (totally agree). We have a similar issue as before that the highest value does correspond to the least gender equal attitude for plh0298_v2 and plh0302_v2. Hence, we recode these as well:

recode plh0298_v2 (1 = 7) (2 = 6) (3 = 5) (5 = 3) (6 = 2) (7 = 1)

recode plh0302_v2 (1 = 7) (2 = 6) (3 = 5) (5 = 3) (6 = 2) (7 = 1)

Then, we add them together into an index by dividing the values by the maximum number of 21. Again, this index ranges from 0 to 1 with 1 being the maximum of 1 referring to the most gender equal attitudes.

Independent variables:

Our main explanatory variable is an indicator of having at least one daughter (information can be obtained from the SOEP using the kidsexXX variable).

Additionally, we control for the number of children (sumkids) and whether they live in the same household (hid).

To control for parental characteristics, we plan to include a dummy variable for sex (sex), age (age), employment status (pgemplst or e11103) and education (pgbilzeit or d11108), as well as place of residence (sampreg) and migration background (migback).

Inclusion/exclusion criteria:

We use data from 2012 and 2018, since attitudes are only available for these two years. We generally exclude respondents without children or with children older than 18 years (age can be calculated by kidgebXX). Additionally, we drop all individuals with missing information on the gender attitude variables or with missing values for the basic information on the children in the household (sex of the child, birth year, number of children). We use the fact that the gender of a child is exogenous (conditional on the number of children) for identification and compare individuals who have at least one daughter to those that have only sons, conditional on the number of children (similar to Borrell-Porta et al. 2019).

Exact analysis:

In the first column, we use the following pooled OLS-model:

$$y_{it} = \propto + \beta_1 daughter_{it} + \beta_2 numchild_{it} + \beta_3 X_{it} + \lambda Survey_t + \varepsilon_{it}$$

In all regressions, $y_{i,t}$ stands for the level of gender equal attitudes (index described above) of individual i in year t. $daughter_i$ is our key regressor for parenting daughters and takes the value of one if individual i has at least one daughter in the household in year t, and zero otherwise. $numchild_{i,t}$ reflects a control variable for the total number of children for individual i in year t.

$X_{i,t}$ is a vector of individual-level control variables including sex, age, employment status, education, place of residence, living with the children, and migration background,

all measured in year t. We additionally include a control für the survey year ($Survey_t$). $\varepsilon_{i,t}$ is the error term. We cluster the standard errors on the household level.

Interpretation of results:

β_1 reflects the effect of having at least one daughter compared to having only sons on gender equal attitudes. In this and all subsequent regressions, we use a 95% confidence interval as the significance threshold, interpreting the effect of having at least one daughter as statistically significant if the corresponding p-value is less than or equal to 0.05.

Hypothesis 2. Having daughters leads to more left-leaning political preferences

That is, denoting Y as a measure of left-leaning political preferences,
(Y for respondents with daughters) > (Y for respondents without daughters)

Dependent variables:

We create a measure for party leanings (plh0012_h), the intensity of party leanings (plh0013_h) and the combination of the two. First, we recode the party leanings variable into a dummy for whether individuals are leaning towards a party right or left of the center where we classify the parties SPD, Greens, the Left, Piratenpartei as left of center and CDU/CSU, AfD, FDP, DVU, Rep, NPD as right of center. Some individuals also support coalitions that go across the center line. We replace their leanings with missing. This means, we end up with the following recoding:

recode plh0012_h (1 5 6 9 16 17 26 = 1) (2 3 4 7 13 14 21 22 25 27 30 = 0) (8 10 11 12 15 18 19 20 23 24 28 29 31 = .)

Second, we use the variable plh0013_h to measure effects on the intensity of party preferences (recoding all values <0 as missings). Finally, we construct a categorical variable that captures whether an individual is intensely left-leaning (4), less intense left-leaning (3), less intensely right-leaning (2) or intensely right-leaning (1) where we utilize the intensity of the party leaning (plh0013_h) and the recoded variable plh0012_h

gen intensity_leaning = .

replace intensity_leaning = 1 if plh0012_h == 0 & inlist(plh0013_h, 1, 2)

replace intensity_leaning = 2 if plh0012_h == 0 & inlist(plh0013_h, 3, 4, 5)

replace intensity_leaning = 3 if plh0012_h == 1 & inlist(plh0013_h, 3, 4, 5)

replace intensity_leaning = 4 if plh0012_h == 1 & inlist(plh0013_h, 1, 2)

In our regressions, we begin by examining the effects on party leanings, followed by political intensity, and finally use the combined measure "leaning intensity" as the outcome variable.

Independent variables:

Our main explanatory variables are the event-time dummies for the arrival of the first child and the interaction with sex of the child. Additionally, we control for the number of children (sumkids).

To control for parental characteristics, we plan to include a dummy variable for sex (sex), age (age), employment status (pgemplst or e11103) and education (pgbilzeit or d11108), as well as place of residence (sampreg), living with the children (hid) and migration background (migback).

Inclusion/exclusion criteria:

We use data from all years with complete outcome variable information, starting in 1995 to avoid potential bias from German reunification. Our sample consists of respondents who eventually become parents in this period. We exclude individuals with missing values for the basic information on the children in the household (sex, birth year, number of children). We use the fact that the gender of a child is exogenous for identification and compare individuals who become the parent of a daughter to those that become parent of a son.

- **Exact analysis**:

We estimate an event-study regression with year FE:

$$y_{i,t} = \lambda_t + \sum_{\substack{k=-k, \\ k \neq -1}}^{K} \delta_k \; FirstBirth_{i,t} + \sum_{\substack{k=-k \\ k \neq -1}}^{K} \beta_k \; (FirstBirth_{i,t} x \; ChildFem) + \gamma X_{i,t} + \varepsilon_{i,t}$$

For each individual in the data set, we define the year they have their first child as event time t = 0, and index all other years relative to this event. Our model uses a unbalanced panel, estimating effects for 10 years before to 10 years after the birth of their first child, so event time k ranges from −10 to +10. The coefficient of interest is β_k as it shows the effect of having a first-time child which is female compared to one which is male.

The outcome $y_{i,t}$ refers to the above-mentioned binary party leaning variable, the intensity variable, and the combined leaning intensity variable. Finally, we include year fixed effects to capture common time trends. Additionally, we include parental specific controls in $X_{i,t}$ (see independent variables). We use robust standard errors.

- **Interpretation of results**:

β_k reflects the coefficient of interest for the different event times. Any significant coefficient before event-time -1 would reflect a pre-trend, meaning that political preferences evolve differently for parents to be of daughters compared to parents to be of sons – we hope that this is not the case. After year -1, significant coefficients β_k indicate significant effects of having a daughter compared to a son on political preferences – having the 10 years after birth further allows us investigate long-term effects of having a daughter. We again interpret coefficients with corresponding p-values <=0.05 as significant.

Hypothesis 3. Having daughters leads to more gender equality in the household (as measured by behaviors or outcomes).

That is, denoting Z as a measure of gender equality in the household,
(Z for respondents with daughters) > (Z for respondents without daughters).

Dependent variables:

We create an index of gender equality outcomes in the household: For this, we use the variables the actual weekly working hours (plb0186_h, non-working parents are coded with 0 hours), weekly care hours for a child (pli0044_h), and weekly hours spent on housework (pli0043_h). First, we calculate the index by creating the share of the mother's weekly hours spent on paid work, and unpaid work, reflected by childcare and housework (share is calculated by hours mother/(sum of hours of mother and father). Since a higher share in unpaid work, reflects more traditional gender roles in

contrast to a higher share in paid work, we recode the share in unpaid work as 1- share unpaid work. From these two shares, we calculate the index, by calculating the row mean of the shares.

Note that gender equality in these outcomes is reached if the index is at 0.5. If it is 0.5 > Index >= 1 than we observe the opposite of traditional gender roles and if it is 0 > Index > 0.5, we observe traditional gender roles.

Independent variables:

Our main explanatory variables are the event-time dummies for the arrival of the first child and the interaction with sex of the child. Additionally, we control for the number of children (sumkids).

To control for parental characteristics, we plan to include a dummy variable for sex (sex), age (age), employment status (pgemplst or e11103) and education (pgbilzeit or d11108), as well as place of residence (sampreg), living with the children (hid) and migration background (migback).

Inclusion/exclusion criteria:

We use data from all years with complete outcome variable information, starting in 1995 to avoid potential bias from German reunification. Our sample consists of respondents who eventually become parents in this period. We exclude individuals with missing values for the basic information on the children in the household (sex, birth year, number of children). We use the fact that the gender of a child is exogenous for identification and compare individuals who become the parent of a daughter to those that become parent of a son.

Exact analysis:

We estimate an event-study regression with year FE:

$$y_{i,t} = \lambda_t + \sum_{k=-k}^{K} \delta_k \; FirstBirth_{i,t} + \sum_{k=-k}^{K} \beta_k \; (FirstBirth_{i,t} x \; ChildFem) + \gamma X_{i,t} + \varepsilon_{i,t}$$

For each individual in the data, we define the year they have their first child as event time t = 0, and index all other years relative to this event. Our model uses a unbalanced panel, estimating effects for 10 years before to 10 years after the birth of their first child,

so event time k ranges from −10 to +10. The coefficient of interest is β_k as it shows the effect of having a first time child which is female compared to one which is male.

The outcome $y_{i,t}$ refers to the above gender equality measure, reflecting the index of shares of paid and unpaid work done by the mother. Finally, we include year fixed effects to capture common time trends. Additionally, we include parental specific controls in $X_{i,t}$ (see independent variables). We use robust standard errors.

Interpretation of results:

β_k reflects the coefficient of interest for the different event times. Any significant coefficient before event-time -1 would reflect a pre-trend, meaning that gender equality evolves differently for parents to be of daughters compared to parents to be of sons – we hope that this is not the case. After year -1, significant coefficients β_k indicate significant effects of having a daughter compared to a son on gender equality measured by the index reflecting the intra-household shares of mothers paid and unpaid work – having the 10 years after birth further allows us investigate long-term effects of having a daughter. We again interpret coefficients with corresponding p-values <=0.05 as significant.

Hypothesis 4. Having daughters leads to a larger increase in the support of gender equality for men than for women (with gender equality measured by attitudes).

That is, denoting X as a measure of support for gender equality,
(X for male respondents with daughters − X for male respondents without daughters) > (X for female respondents with daughters − X for female respondents without daughters)

- **Variables**:

Dependent variables:

As an outcome variable, we propose to measure effects on two indices of gender equality attitudes (one for 2012 and one for 2018). The index should be composed of variables plh0298_v1, plh0302_v1, plh0308_v1 for 2012 and plh0298_v2, plh0302_v2, plh0308_v2 for 2018. The variables that end in _v1 are on a scale of 1 (fully agree) to 4

(do not agree) with a maximum of gender equal attitudes at 4 for the variables plh298_v1 and plh0302_v1. The variable plh0308_v1 has the most gender equal attitude at 1. So, in a first step, we recode this variable in a way that 4 also marks the most gender equal attitude:

recode plh0308_v1 (1 = 4) (2 = 3) (3 = 2) (4 = 1)

In the next step, we add the values from all three variables and divide them by 12 (the maximum sum) to create a score for gender equality attitudes that ranges from 0 to 1 with 1 being the maximum of gender equality attitudes.

Additionally, an index for gender equality in 2018 is our second outcome variable. For 2018, the variables that end on _v2 are on a scale of 1 (do not agree at all) to 7 (totally agree). We have a similar issue as before that the highest value does correspond to the least gender equal attitude for plh0298_v2 and plh0302_v2. Hence, we recode these as well:

recode plh0298_v2 (1 = 7) (2 = 6) (3 = 5) (5 = 3) (6 = 2) (7 = 1)

recode plh0302_v2 (1 = 7) (2 = 6) (3 = 5) (5 = 3) (6 = 2) (7 = 1)

Then, we add them together into an index by dividing the values by the maximum number of 21. Again, this index ranges from 0 to 1 with 1 being the maximum of 1 referring to the most gender equal attitudes.

Independent variables:

Our main explanatory variable is an interaction term of the indicator of having at least one daughter (information can be obtained in the SOEP with the variable kidsexXX) with the sex of the parent (variable sex in the SOEP).

Additionally, we want to control for the number of children/sons/daughters (sumkids, kidsexXX, kidgebXX) and whether they live in the same household (hid).

In order to control for parental charatceristics, we plant to include a dummy for sex (sex), age (age), occupational position (pgstib e11105)/employment status (pgemplst or e11103)/education (pgbilzeit or d11108), place of residence (sampreg), migration background (migback).

Inclusion/exclusion criteria:

We use data from 2012 and 2018, since attitudes are only available for these two years. We generally exclude respondents without children or with children older than 18 years (age can be calculated by kidgebXX). Additionally, we drop all individuals with missing information on the gender attitude variables, with missing values for the sex of the parent, or with missing values for the basic information on the children in the household (sex of the child, birth year, number of children). We use the fact that the gender of a child is exogenous (conditional on the number of children) for identification and compare men and women who have at least one daughter to those that have only sons, conditional on the number of children (similar to Borrell-Porta et al. 2019).

Exact analysis:

We use the following pooled OLS-model:

$$y_{it} = \propto + \beta_1 daughter_{it} + \beta_2 (daughter_{it} x\ woman_i) + \beta_3 woman_{it} + \beta_4 numchild_{it} + \beta_5 X_{it} + \lambda Survey_t + \varepsilon_{it}$$

In all regressions, $y_{i,t}$ stands for the level of gender equal attitudes (index described above) of individual i in year t. $daughter_{it}$ indicates having at least one daughter, $woman_{it}$ indicates that the respondent is female. β_2 is our coefficient of interest and shows the heterogeneous effect on attitudes for women compared to men. $numchild_{i,t}$ reflects a control variable for the total number of children for individual i in year t.

$X_{i,t}$ is a vector of individual-level control variables including age, employment status, education, place of residence, living with the children, and migration background, all measured in year t. We additionally include a control für the survey year ($Survey_t$). $\varepsilon_{i,t}$ is the error term. We cluster the standard errors on the household level.

Interpretation of results:

We interpret β_2 as the difference in the effect on gender equal attitudes of having at least one daughter compared to having only sons for women in comparison to men. We use a 95% confidence interval as the significance threshold, interpreting the effect of having at least one daughter as statistically significant if the corresponding p-value is less than or equal to 0.05.